

Fear of Temperature

Research direction and initial source-feasibility progress

Computational analysis of policy, media and public climate emotions

Government / policy	News media	Public expression
COMPLETED · bounded M1.1 audit	PLANNED · NLP + temporal analysis	DECISION · counting / text / rights

One-minute overview

The topic has not changed: how policy, media and public communication express fear of rising temperatures, who changes first, what changes around events and how responsibility is explained. The method moves from keyword/Ngram exploration towards a traceable corpus, validated semantic retrieval and bounded temporal analysis. Lexical and TF-IDF methods remain baselines; 1988 is an IPCC-related collection anchor, not an emotional origin.

Completed this week: nine July 2026 GOV.UK policy_paper records tagged to DEFRA. Reported/returned: 9/9; pagination: 5 + 4; accessible API records and pages: 9/9; missing and duplicate key fields: 0. This supports one bounded document denominator only. Retained bodies: 0; no emotion or temporal result exists.

Dai Pan · Supervisor: Mashhuda Glencross · progress report, not a completed model study

RESEARCH TRANSITION

The research question is unchanged; the evidence strategy is changing

Earlier exploration	Current design	Reason
Dictionaries, keywords, Ngrams	Traceable corpus + semantic retrieval	Similar concerns use different language; the same word can be quotation, denial or unrelated text
Raw hit counts	S, E and derived B with aligned denominators	A hit count is not attention unless numerator and denominator share the eligible collection
Selected examples	Versioned queries, held-out validation, evidence spans	Reduces discretionary selection and preserves counterexamples
General negative sentiment	Emotion, target, horizon, holder, cause, blame and duty	Similarity is not intensity; a risk statement need not express fear

Baseline retained

Dictionary and TF-IDF methods remain. A Transformer only counts as an improvement if held-out precision, recall, F1 and error analysis support it.

1988 boundary

The IPCC provides an institutional collection anchor, not the origin of fear or a demonstrated breakpoint. The final comparison window depends on reliable overlap.

Historical span is not comparable coverage

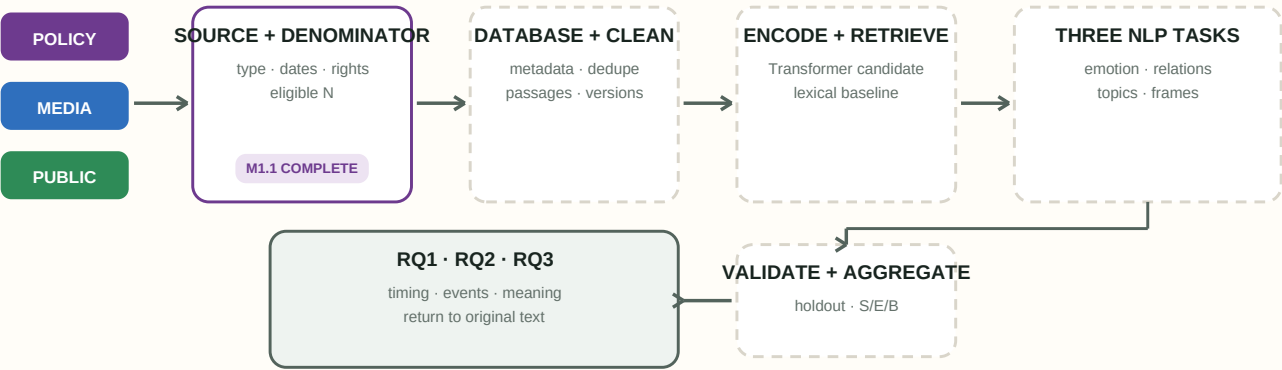
Letters, early forums and modern platforms will not be merged into a seamless public-emotion series. Channel changes can flag composition risk but cannot prove bias by themselves.

The scholarly question remains social: the corpus and code make the evidence reproducible; they are not the substantive contribution by themselves.

END-TO-END EVIDENCE CHAIN

Original evidence remains linked throughout the planned workflow

Design diagram: dashed stages are planned; only M1.1 source feasibility is completed



Evidence spine: document → passage → date/source → speaker/holder → span → model version → aggregate → interpretation

Measures	S: relevant share of eligible units E: future fear/worry within relevant text B = S x E: derived joint share
Model boundary	Sentence-BERT is a retrieval candidate; GoEmotions is a dataset; spaCy/Stanza is not a complete responsibility system.
Evidence boundary	Publishing institution, quoted speaker and emotion holder remain separate. Evidence spans, negation, versions and counterexamples are retained.

Planned, not completed: exact checkpoints are not frozen; shared representation does not imply joint training or algorithmic novelty. Aggregation follows source/period validation.

RESEARCH QUESTIONS

Three RQs connect timing, events and meaning

RQ	Question	Planned evidence and method	Boundary
RQ1 Who changes first?	Leading, lagging, synchronous and feedback patterns across roles	Common-window S/E; CCF; conditional VAR against own-history baseline	Precedence and added prediction are not automatic causation
RQ2 What changes around events?	Level and slope near independent heat, science or policy dates	Segmented regression / ITS; windows; placebo dates; serial-dependence checks	Change may reflect anticipation, common shocks, composition or channel shifts
RQ3 How is fear explained?	Affected groups, causes, blame, duties and threatened futures	Evidence-linked relations; topics/frames; quotation and stance; contrary text	Grammar is not culpability; similarity is not agreement or intensity

Why all three?

RQ1 without RQ3 gives curves without meaning; RQ3 alone cannot compare sequence; RQ2 links independently dated events to changes. NLP components serve multiple RQs rather than mapping one model to one question.

Shared causal boundary

CCF/Granger/VAR added prediction is not causal identification. A non-significant result would not prove policy dominance or the absence of physical influence. Channel-transition diagnostics only flag composition risk.

COMPLETED THIS WEEK

M1.1 verified one bounded policy-document denominator

Actual access chain: each full bar is 9/9



First published	Real title	Why it matters
3 Jul 2026	Air Pollution Awareness Coalition (APAC)	Updated 21 Aug; DEFRA, DHSC and DfT. Publication and update dates differ.
13 Jul 2026	30by30 on land in England: Delivery plan	policy_paper; linked to DEFRA.
15 Jul 2026	British Sign Language 5-year plan: Department for Environment, Food and Rural Affairs - 1-year update, July 2026	Non-climate record: denominator was not built from climate keyword hits.

Decision: status 1, conditional go

N_total = 9 is complete only for this saved index time, organisation tag, document type and month. It is not all DEFRA output, all UK policy, a passage denominator or proof of complete 1988-2026 coverage.

TECHNICAL FINDINGS AND LIMITS

What the pilot established - and what remains unclaimed

Completed and confirmed

- `first_published_at` worked as a filter, but was not populated in the search response; sorting returned HTTP 422.
- The Content API supplied dates for verification and local sorting.
- 3/9 records had multiple organisations; the pilot counted each landing page once if DEFRA-tagged.
- Access time and evidence hashes were saved.

Technical implications

- `public_timestamp` can reflect an update and cannot substitute for first publication.
- An organisation tag is not exclusive authorship; cross-institution totals may double count.
- The index is a dynamic snapshot.
- One complete month does not prove 1988-2026 completeness.

Not completed

Body/attachment retention, passage denominator, relevance/emotion labels, embeddings, topics, relations, S/E/B, CCF, VAR and ITS. Retained bodies and paragraph examples: 0.

Rights and ethics

Most GOV.UK material is under OGL v3, but attachment, third-party and personal-data exceptions remain. Public visibility is not blanket permission to store or redistribute content.

The pilot cannot establish: climate prevalence, fear level or trend, policy leadership, whole-government representativeness or complete historical coverage.

NEXT WEEK AND DECISIONS

21-27 September: 10-20 hours, strengthen measurement before expansion

Time	Work	Output / stop condition
3-5 h	Preserve snapshot; audit recent, intermediate and earlier windows	Versioned coverage table; do not overwrite M1.1; downgrade status if fields are unstable
2-3 h	Review three multi-organisation records and date/unit rules	Data dictionary and counting recommendation; no cross-institution total before decision
2-4 h	If authorised, test provenance on three public documents	Traceable passage sample; otherwise metadata or labelled synthetic checks only
3-5 h	Two close readings + two targeted readings	Query and annotation drafts; readings are planned, not completed
0-3 h	Synthesis and demonstration	Next meeting material; total stays within 20 h

Planned reading

Close: Sentence-BERT; Pihkala climate-emotion taxonomy.
Targeted: GoEmotions transfer limits; Brulle et al. temporal design.

Supervisor decisions requested

- 1988 anchor with final analysis restricted to common coverage?
- Joint records: all linked, lead-only or fractional weights?
- What supervision, ethics and rights record is needed before a three-document passage pilot?
- Prioritise coverage audit or freeze units/counting first?

Closing principle

Make the denominator, evidence location and validation chain defensible before running complex models. Next week does not commit to completing emotion models or VAR.

Evidence base: meeting_report_en.md; latest proposal; research narrative; saved M1.1 README, CSV, denominator assessment and checks.json. This report did not rerun collection.